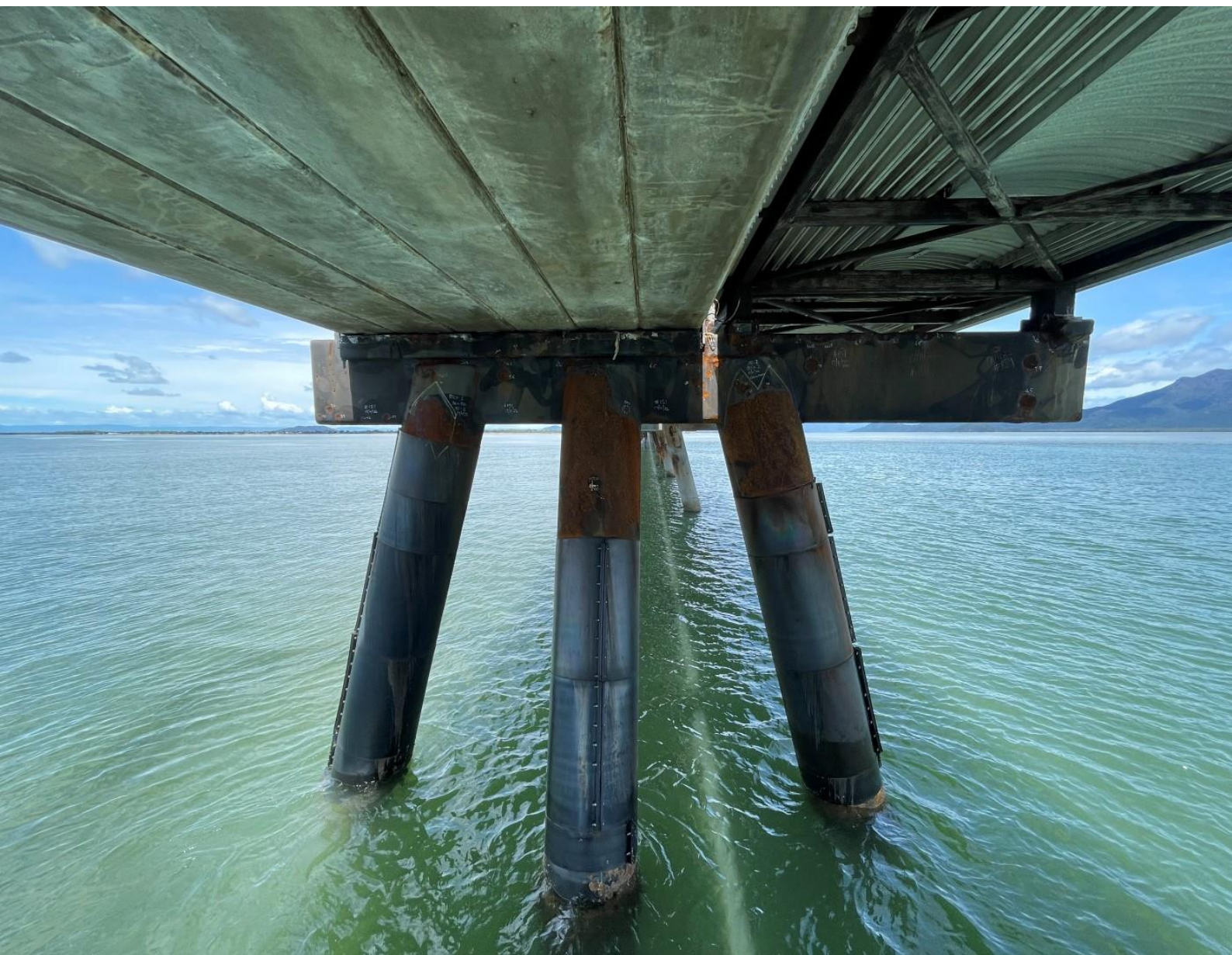


Qld Sugar Limited

Lucinda Bulk Sugar Terminal (LBST)

Phase 1 Summary Report | Modelling of Maintenance Strategy Costs of the LBST Substructure

SCMC-22020-REP-001 | Revision 1 | 20 March 2023



Phase 1 Summary Report: Lucinda Bulk Sugar Terminal (LBST)

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Appendix A SCMC Review Comment Register

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1 Introduction

In November 2022, Shoreline Civil and Marine Consulting (SCMC) was engaged by Queensland Sugar Limited (QSL) to undertake 'Modelling of Maintenance Strategy Costs' for the Lucinda Bulk Sugar Terminal (LBST) substructure.

We understand that the overall objective of the project is to develop a financial model that will allow the application of different maintenance strategies to determine / evaluate the outcome in terms of ensuring the structural integrity and safety of the asset is maintained and estimating the forward maintenance costs for the LBST (jetty and wharf) piles and headstocks. We understand the financial model will be used to assist QSL in the forward works planning for capex and major maintenance.

1.1 Project Understanding and Background

Based on our previous experience from inspecting and reporting on the condition of LBST in early 2022 and from discussions with QSL and review of historical documentation, we understand the project background and context to be as follows:

1. LBST wharf and jetty was originally constructed around 1977-1979 (the current age of the original wharf, dolphin and jetty components is 44 - 46 years). It is understood that major structural repairs were undertaken in 2012 (designed by Aurecon) following widespread damage to the facility caused by Tropical Cyclone Yasi in February 2011.
2. As shown below in Figure 1, the LBST wharf and jetty comprises of the following areas / zones:
 - **Jetty**
 - **Wharf:** Northern Mooring Dolphin, Transfer Tower, Northern Strongpoint Platform, Wharf Roadway, Southern Platform, Southern Mooring Dolphin

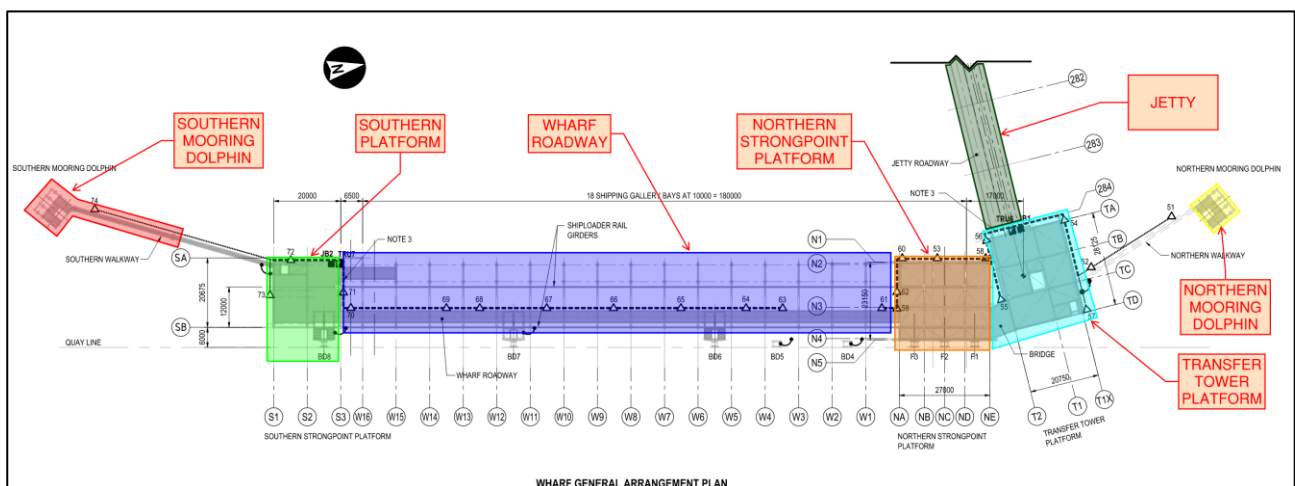


Figure 1: Wharf Areas (markup of Aurecon (2012) Drawing CP-005)

3. The 5.7km approx. long jetty is supported by 284 headstock / pile bents and 670 piles (refer Figure 2) and the wharf head is supported by 153 headstocks and 159 piles (refer Figure 3). These headstocks and piles comprise closed hollow section steel components protected by various paint coating systems adopted over its life to date.
4. Major maintenance expenditure on the Lucinda wharf was resumed circa 2010 after an extended hiatus of approximately 5 years where the maintenance budget on this asset had been significantly reduced.

5. During the last 10 years, QSL has changed the pile corrosion protection system from a paint coating protective system to a petrolatum tape and polymer jacket system (Denso SeaShield 2000 FD system or equivalent) which according to QSL has dramatically increased the durability protection for the piles and have also trialled a number of different protection systems for the headstocks.
6. The Jetty Maintenance Vehicle (JMV) was also upgraded to enable the installation of the new pile wrap protection system dramatically reducing the installation / access costs.
7. Repairs in the last 10 years have predominantly focussed on the wrapping of piles rather than the improvement of the condition of the head stocks as the paint protection on these was still in mostly in good condition. However, the headstock protection systems are now starting to deteriorate significantly, and the maintenance of the headstocks is now necessary, or significant additional costs for corrosion damage rectification will be incurred in the coming years.



Figure 2: LBST Jetty Substructure Arrangement (Typical)



Figure 3: LBST Wharf Substructure Arrangement (Northern Strongpoint Repaired in 2021)

1.2 Project Objectives

Noting the above project background and context, we understand that:

- the LBST jetty pile maintenance program has reached the point of having wrapped all the piles on the jetty that do not require weld / strengthening repairs;
- QSL has concerns that it may not be possible to “keep ahead” of the maintenance of the remaining wharf and jetty piles with the resources (budget and equipment) currently available;
- this question has been highlighted by the noted significant accelerated deterioration of piles that require weld repairs and the significant drop off in piles wrapped per year due to the impact of first needing to conduct costly and time intensive weld / strengthening repairs;
- the headstock protection systems are now starting to deteriorate significantly, and the maintenance of the headstocks is now necessary, or significant additional costs for corrosion damage rectification will be incurred in the coming years;
- there is also the further consideration of what impacts additional early expenditure may have in reducing the total maintenance cost of the wharf over the next 25 years, presenting a potential cost saving for QSL.

Accordingly, we understand that the overall objectives of this project is to:

1. Review all relevant historical documentation in relation to maintenance and repair of the LBST steel piles and headstocks;
2. Engage with relevant stakeholders to provide input and feedback into this review;

3. Develop a financial model that will allow the application of different maintenance strategies to determine / evaluate the outcome in terms of ensuring the structural integrity and safety of the asset is maintained and estimating the forward maintenance costs for the LBST (jetty and wharf) piles and headstocks. We understand the financial model will be used to assist QSL in the forward works planning for capex and major maintenance.

1.3 Project Scope

As outlined in SCMC's proposal dated 17th November 2022, the scope of services to be undertaken as part of this project includes:

1. Phase 1 - Asset Review and Reporting:

- Project kick-off meeting with the QSL project team; and
- Detailed review of all relevant previous documentation as provided and listed in Table 1 of the QSL RFP brief (refer Section 2 of this report for details / outcomes of this review).
- Develop recommended pile and headstock maintenance strategy options over the next 25 years for discussion / brainstorming at a workshop with QSL personnel and their jetty pile maintenance contractor (Advanced Aqua Blasting (AAB)).
- Facilitate a workshop with QSL and AAB to gain an understanding of the required inputs needed for preparation of the Financial Model including typical repair production rates, cost rates, challenges faced, issues experienced with current repair procedure, etc.
- Discussion / exploration of possible options for accelerating maintenance repairs to the LBST piles and headstocks.
- Prepare a Phase 1 Summary Report (this report) documenting the findings / outcomes of Phase 1 and to provide the required input for the development of Phase 2 outputs as outlined below.

2. Phase 2 - Development of Financial Model:

- Compilation of a financial model based on the findings and outcomes of Phase 1 to assess repair and maintenance options for the next 25 years for each of the piles and headstocks of the jetty and wharf.
- The model will be compiled to provide a direct comparison between the various agreed maintenance strategy options (as determined in Phase 1) to determine which strategy will have the best benefit / return on investment for QSL. This will be assessed using a net present value analysis of assumed capital expenditure outputs for each maintenance strategy option over the 25 year time period.
- Update the Phase 1 Summary Report to include the Phase 2 financial model assessment findings / outcomes.

1.4 Scope Limitations

The scope of this project relates only to pile wrapping, protective coating system and section loss repairs to the piles and headstocks for the LBST steel headstocks, below deck steel bracing and steel piles down to mean sea level. For the avoidance of doubt, the scope of this project does not include:

- Below water elements including the cathodic protection system
- Concrete girders and steel conveyor structure (above headstocks)
- Wharf furniture, fenders, fender panels and chains
- Mechanical equipment
- Access walkways underneath jetty and all above deck elements

2 Documentation Review and Discussion

SCMC have undertaken a detailed review of all relevant existing LBST documentation, as provided by QSL. The list of documents reviewed is summarised in Table 1.

A detailed record of SCMC's review can be found in the comment register provided in **Appendix A**.

Table 1: LBST Documentation provided to SCMC for Review

Project Name / Timeline	Description	Year	Document Details
Commissioning of Asset	Piles and headstocks were initially shop coated in July 1977 at Steel Mains in Townsville with a coal tar epoxy coating system from RL -2 metres to cut-off. The system consisted of a blast cleaning to Sa3 standard with a surface profile of 40 microns followed by two coats of Dimet Tarsel coal tar epoxy to give a minimum dry film thickness of 400 microns. The original paint record sheets showed values of at least 500 microns thickness were achieved. Damaged regions were touched up once installation was finalised.	1979	Reference Document #01 - Original Construction Plans
MacDonald Wagner Paint Investigation Report	MacDonald Wagner carried out an investigation of coating condition in 1986 (Paint Investigation Report, 1986) looking at erosion and other damage to the coal tar epoxy coating. The investigation found that, after eight years exposure: - Erosion of the coating on the jetty and wharf piling was approximately 10 microns per year on average, with some regions of 20 microns or more per year - Blistering in the tidal zone was observed on only two of 250 piles examined - Disbondment of paint within the barnacle zone was observed on one half of the 250 piles examined. There was no relationship between disbondment and location of CP anodes - Blistering and disbondment were not considered to have been due to excessive cathodic protection The degree of erosion was considered normal for a coating of this type. The disbondment in the tidal zone was of concern and a long-term maintenance program was recommended to be formulated to ensure that disbonded paint areas not protected by cathodic protection (above mid tide level) were quickly repaired.	1986	Reference Document #02
Expansion Joint Maintenance	Maintenance to 6 of the 20 expansion joint headstocks was undertaken. It is understood that the concrete deck units were jacked up to provide access and the top surface of the headstocks were repaired using a stainless steel plate with "Flurogold" bearings bonded onto it, fully seal welded onto the top of the headstock.	1998	Information pack provided by QSL on 17/02/2023

Project Name / Timeline	Description	Year	Document Details
Maintenance Hiatus	Exact period is unknown. Discussions with long term QSL staff indicate that it was for approximately 5 years.	2005 – 2010 (approx.)	N/A
Aurecon Corrosion Protection of Steelwork	QSL requested Aurecon to review the protective coating systems used to protect the steel piles and headstocks on the jetty and offshore structure. Specifically, Aurecon was requested to: a) Identify and review coating systems currently available on the market against the QSL preferred International Paints Interzone 954 system currently adopted b) Recommend a corrosion protection coating system to be taken forward under the new five-year corrosion protection program for QSL review and approval c) Recommend a corrosion protection maintenance program d) Provide a detailed technical specification of the approved system suitable for inclusion in a Contract for procurement of external specialist contractors	Jun 2009	Reference Document #03
Pile Inspection Reports	Pile inspection reports with photos.	Dec 2009	Reference Document #04
Aurecon Investigation into Structural Integrity Lucinda BST Pile and Headstock Corrosion	In March 2010, Queensland Sugar Limited (QSL) commissioned Aurecon to undertake the following study regarding the corroded piles and headstocks on the Lucinda jetty: a) Review the International condition assessment report findings specifically the “black priority” pile bents nominated in the report b) Inspect the area of pile and headstock walls that is corroded & a reduced wall thickness. c) Confirm effect, if any, on the structural integrity of the pile and headstock d) Identify remedial works, if any, to be undertaken to ensure structural integrity of the pile and headstock e) Undertake cost comparison and whole of life costing comparison of the Denso Seashield system compared with the International Interzone 954 coating system for the case where the pile needs complete recoating	Apr 2010	Reference Document #05
Jetty Pile Tracker	Tracking of maintenance works on jetty and wharf piles. Spreadsheet has been tracking since 2010 collating data to date	2010 – to date	Reference Document #06
Aurecon 2010 Site Inspections Summary Report - Pile Repair Works	Aurecon undertook periodic site inspections during rectification works on 39 pile bents. This report provides a summary of the site inspections for the repair works in 2010.	Apr 2011	Reference Document #07

Project Name / Timeline	Description	Year	Document Details
Aurecon Pile Utilisation Assessment	A structural assessment of the various pile bents was undertaken with the aim to optimise or eliminate steel repairs and to aid in prioritising corrosion protection works.	2013	Reference Document #08
Commonline Condition Assessment	Corrosion survey report to provide data on the current condition of the Lucinda Jetty and Wharf Pile and Headstock structures.	2016	https://qsl.com-monline.com.au/ Username: sugar Password: pilespostsbeams
Pile Maintenance Records	Corrosion damage logs from the pile maintenance.	2018 – to date	Reference Document #09
TAMS Group Dive Inspections	TAMS Group was engaged by Queensland Sugar Limited (QSL) to conduct diving inspection works on QSL Bulk Sugar Terminal Wharf in Lucinda. Ultra-Thickness readings, Cathodic Protection readings and visual inspection were conducted and reported.	Sep 2020	Reference Document #10
Pile Repair Rate Analysis	A sample of weld repairs in 2021 were documented to calculate the repair rate based on \$/m2. It tracked the welding hours vs size of patch to determine the cost. This can be expanded based on more recent progress claims in a similar manner.	Nov 2021	Reference Document #11
Shoreline Condition Assessment Report	This report details the condition assessment findings from the Ports Australia WSCAM Basic Visual inspection of the Lucinda Bulk Sugar Terminal (LBST). The condition assessment of the LBST was undertaken by SCMC between the 18th and 25th January 2022 and included visual assessment of the structure by foot (above deck) and by boat (below deck) above the water line. No diving inspection was undertaken as part of this assessment.	Apr 2022	Reference Document #12
Shoreline Panoramic Models	A full panoramic photo survey is available online of the jetty and wharf structure.	Apr 2022	Login page: https://shore-linecmc.net/login/ Email: qsl@shore-linecmc.com.au Password: qslbstinspections2022

A summary of the key review findings relevant specifically to the Phase 1 and Phase 2 deliverables are as follows:

- **Pile Repair Methodology (Aurecon Pile Utilisation Assessment - Reference Document #08):**

- Aurecon confirmed that there was no accompanying report for the current weld repair methodology and utilisation assessment (developed by Aurecon in 2012 / 2013). The design criteria, calculation approach, key assumptions and critical load cases are therefore unknown.
- The original wall thickness of the piles is generally 10mm, however the repair plates are 8mm thick with 8mm fillet welds noting that the 8mm fillet welds only have a structural throat thickness of ~5.6mm which is significantly less than the original pile thickness. It therefore appears that the design intent is to partially rely on the residual wall thickness.
- It is understood that QSL's jetty maintenance contractor (AAB) has recently modified the pile weld repair procedure due to increased corrosion section loss areas on the piles and to reduce the number and associated cost of individual weld repairs. The new procedure involves welding full circumference 8mm collar plates (in 4 quartered 1m high sections) over the corroded zones of the piles.

- **2022 SCMC Condition Assessment vs Pile Tracker (Reference Document #06 and #12):**

- The 2022 WSCAM condition assessment was performed by SCMC in January 2022. It appears from the “Lucinda Pile Tracker V8” spreadsheet that was provided to SCMC that repairs have been performed to 33 additional piles since the 2022 SCMC condition assessment and is therefore more up to date than the 2022 SCMC condition assessment schedule. However, in our review we did find a number of discrepancies between the 2022 SCMC condition assessment and the pile tracker spreadsheet. Spot checks of the discrepancies was performed using the SCMC panoramic model and the 2022 SCMC condition assessment data was found to be correct. Therefore, the pile condition from on the 2022 SCMC condition assessment data has been adopted for the purposes of this review / report.

- **Pile Repair Rate Analysis (Reference Document #11):**

- There seems to be a number of conflicting trends within the corrosion rate data and as such no apparent conclusions that can be relied upon from this assessment.

3 Workshop Discussion and Outcomes

3.1 Workshop Minutes

A workshop was held at the Lucinda QSL office on the 2nd February 2023 with QSL personnel and their current jetty pile maintenance contractor (AAB). SCMC presented a background of the project, scope and objectives and a summary of the asset documentation review. An open discussion was held with QSL and AAB regarding the current repair cost rates, challenges faced and time required to repair a pile and headstock based on various degrees of deterioration as well as potential opportunities for accelerating and improving the current repair procedure/s.

The minutes of this workshop are documented within the Powerpoint Presentation slides attached in **Appendix B**. The findings / outcomes from this workshop were used by SCMC to further understand current condition of the jetty and wharf and inform the development of the recommended maintenance strategy options.

A summary of some of the key workshop discussion items is provided below.

3.2 LBST Repair Work Access Systems

Based on the documentation review and discussions during the workshop, it is understood that different access systems are currently used for the various components of the LBST jetty and wharf due to the varying structural forms and existing equipment available.

The Jetty Maintenance Vehicle (JMV – refer Figure 4 and Figure 5) is currently the primary method of access for the jetty pile and headstock repairs. In recent years, the main maintenance activity on the jetty has been pile wrapping and weld repairs. QSL and AAB have advised that the key production rate constraints in the current work repair methodology include the following:

- Repair/s production rate limited by access with only 1 JMV available
- Key JMV issues and limitations:
 - a. Reliability issues (due to age (estimated at ~40years old) and corrosive environment results in issues with control box, mechanical & electrical gear).
 - b. Weight and space limitations means that there is a 2 person limit on the JMV platform and generally only one person is able to work at any one time.
 - c. Full encapsulation is not practical which is required for capture of sandblasting garnet (it is also noted that encapsulation also improves the quality of coating application and can reduce weather downtime).
 - d. JMV is not able to access the full length of the landward side of headstocks due to rotation angle limitation of the JMV platform.
- Multi-stage sequential repair procedure for the piles with different equipment required for each stage (i.e. Blasting / cleaning -> inspection and UT testing -> weld repair -> pile wrap)
- Weld repairs are time intensive / slow production rate, however this has been sped up by recently modifying the plate repair procedure from multiple small plates to larger full circumferential plates
- Working within tidal windows
- Weather conditions limiting available work time

The JMV is generally not suitable for use on the Wharf. The access methodology for the recent ~2021 AAB wharf repair campaign was to assemble a full extent below deck scaffold platforms for the Northern Strong Point.

Based on the documentation review and discussions during the workshop, it is SCMC's opinion that one of the key production constraints with the current repair methodology is the JMV and that alternative access methodologies will be required to speed up the repair production rate. This will also enable multiple work fronts to work simultaneously in parallel.

Various potential improvements to the current repair methodology have been identified and these are discussed further in Section 5.3. However, for the purpose of this assessment and development of the repair strategy options detailed herein, the following access systems and methodologies have been assumed:

1. **Jetty Pile Repairs:** Access is via. the JMV (or potentially via. scaffolding or movable access platform)
2. **Jetty Headstock Repairs:** Scaffolding with full encapsulation (or potentially via. movable access platform)
3. **Jetty Headstock Repairs:** Combination of rope access and mobile scaffold system, but this would not allow for encapsulation and would need to utilise a wrapping system i.e. Stopaq or similar for repair.
4. **Wharf Piles and Headstock Repairs (including dolphins):** Scaffolding (or potentially via. movable access platform in some areas)

For efficiency reasons, it is recommended that maintenance works performed on the wharf are completed in zones with all works performed in a single repair campaign. For example, one campaign could involve assembling a scaffold platform under the full extent of the Transfer Tower to enable all pile and headstock repairs.

It is expected that a significant proportion of the repair cost is the assembly and disassembly of the scaffolding and it is therefore recommended that a full blast and recoat of the headstock beams are performed (as per 2021 NSP repairs vs. only performing patch repairs were required).



Figure 4: LBST Jetty Maintenance Vehicle (JMV, January 2021)



Figure 5: LBST JMV underneath the jetty (JMV, February 2023)

4 LBST Piles and Headstocks - Current Condition Summary

4.1 Introduction

The aim of this section is to summarise the current condition of the LBST jetty and wharf piles and headstocks and provide related discussion on the scope of the expected required maintenance repairs.

Since the original construction of the LBST jetty and wharf in 1977-1979, the steel headstocks and piles have been protected and maintained using a variety of different paint coating protective systems. Since around 2010, QSL has changed the pile corrosion protection system from a paint coating protective system to a petrolatum tape and polymer jacket system (Denso SeaShield 2000 FD system or equivalent).

It is also understood that the pile repairs in the last 10 years or so have been predominantly focussed on the wrapping of piles that do not require weld repairs. Only in more recent years has the repair works been focussed more around weld repairs on the piles and based on discussions in the workshop, the current focus has been on piles that have either been observed to have localized full section loss (i.e. holes) or are 100% utilised. The current weld procedure involves welding doubler plates over the corroded area of the pile that has experienced section loss (Refer to Reference Document #08 for full details).

SCMC has undertaken a detailed review of the available pile and headstock condition data (Reference Documents #06 & #12) and the results are summarised in the Sections below noting the following:

- Where an element was recorded to have a variable condition in the 2022 SCMC WSCAM data, the overall WSCAM rating has been based on the worst rating. (for example: 50% of the element has a WSCAM rating of 3 and 50% has a WSCAM rating of 5, an overall WSCAM rating of 5 has been adopted).
- Except for minor maintenance noted in the 2021 SCMC condition assessment, it has been assumed that no maintenance is required to piles that already have a pile wrap system installed. It should be noted that there is a risk that the pile wrap system could be hiding areas with unacceptable section loss due to historical section loss that occurred before the wrap was applied, and/or corrosion could be occurring underneath the pile wrap due to failure of the wrap system and/or workmanship issues during installation.

4.2 Jetty Piles – Current Condition Summary

Based on the SCMC 2022 WSCAM condition assessment data, of the 670 piles on the jetty:

- 380 no. (~58%) have been fully wrapped¹.
- 57 no. (~8%) have been partially wrapped²
- 233 no. (~35%) have not been wrapped.

Of the 233 piles that have not yet been wrapped:

- 172 no. (~74%) have a WSCAM condition state rating of 5
- 46 no. (~20%) have a WSCAM condition state rating of 4
- 15 no. have a WSCAM condition state rating of 3

Furthermore:

- As approximately 93% of the piles that have not been wrapped have a WSCAM CS rating of 4 or 5, it is likely that most of the piles that have not been wrapped will require some form of weld repair. AAB also anecdotally advised SCMC that all piles that have not yet been wrapped are likely to require a weld repair.
- Table 2 shows the 2022 SCMC inspection estimated area of corrosion on the piles with respect to the observed WSCAM rating which can be used as a high level indication of the expected size / extent of weld repair required. As expected, the size of corrosion area for piles increases as the overall condition of the pile deteriorates.

¹ 17 of the 380 fully wrapped piles require pile wrap repairs due to localized wrap punctures or missing flange bolts.

² According to the pile tracker spreadsheet, 14 of these 57 piles require a weld repair.

Table 2: Number of jetty piles with respect to WSCAM CS Rating and Estimated Corrosion Area

	Estimated Corrosion Area (m ²)													
	0.5	1	2	3	4	5	6	9	10	11	12	13	14	15
WSCAM Rating 3	44	7	1	1	1									
WSCAM Rating 4	109	48	12	11	5	1	3							
WSCAM Rating 5	3	15	21	22	8	1	23	44	1		50			26

It is also noted that there are 74 piles on the QSL pile tracker spreadsheet with status of “PC” (Full blast and paint since 2012). However, 62 of these piles were observed to have at least part of the pile with a WSCAM CS rating of 4 or 5 suggesting that repairs are required again and that weld repairs might even be required on these piles.

4.3 Jetty Headstocks – Current Condition Summary

4.3.1 Non-Expansion Joint Headstocks

Based on the SCMC 2022 WSCAM condition assessment data, Table 3 below shows the number of headstocks within each WSCAM CS rating category.

Of the 285 jetty headstocks, there are only 13 headstocks (~4%) that appear to have been recently recoated with no defects observed and therefore currently not likely to require any maintenance works.

Table 3: Number of jetty headstocks with respect to WSCAM Rating

	1	2	3	4	5	6	7
Number of Headstocks		12	85	124	64		

Based on the data, it is expected that the majority of the headstocks will require some form of plate weld repair. Table 4 shows the estimated corrosion quantity which provides an indication of the expected size / extent of weld repairs that could be required.

Table 4: Number of jetty headstocks with respect to WSCAM CS Ratings and Estimated Corrosion Area

	Estimated Corrosion Area (m²)														
	0.25	0.5	1	1.5	2	2.5	3	4	5	6	7	8	9	10	
WSCAM Rating 3	1	7	8	5	19	16		2		10		16			
WSCAM Rating 4			3	1	12		37	31	19	11		10			
WSCAM Rating 5		1			2		2	19	15	15		9		1	

4.3.2 Expansion Joint Headstocks

There are 26 no. expansion joint (E.J) pile bents located along the LBST jetty. As shown in Figure 6:

- a stainless steel bearing plate and 2 no. fluorogold slide plates are provided to allow for expansion / contraction of the PRC deck units.
- A 3mm thick neoprene pad is bonded to the full width of the headstock with an approved adhesive.
- Due to the required sliding function of the expansion joints, there is a ~125mm ungrouted gap between the deck units and headstock.

- There is no access to the headstock top plate for inspection and/or maintenance and this was deemed a high risk corrosion area in the 2022 SCMC Condition Assessment and AMP.
- To provide access for inspection and maintenance, the concrete deck units need to be lifted (jacked) up.

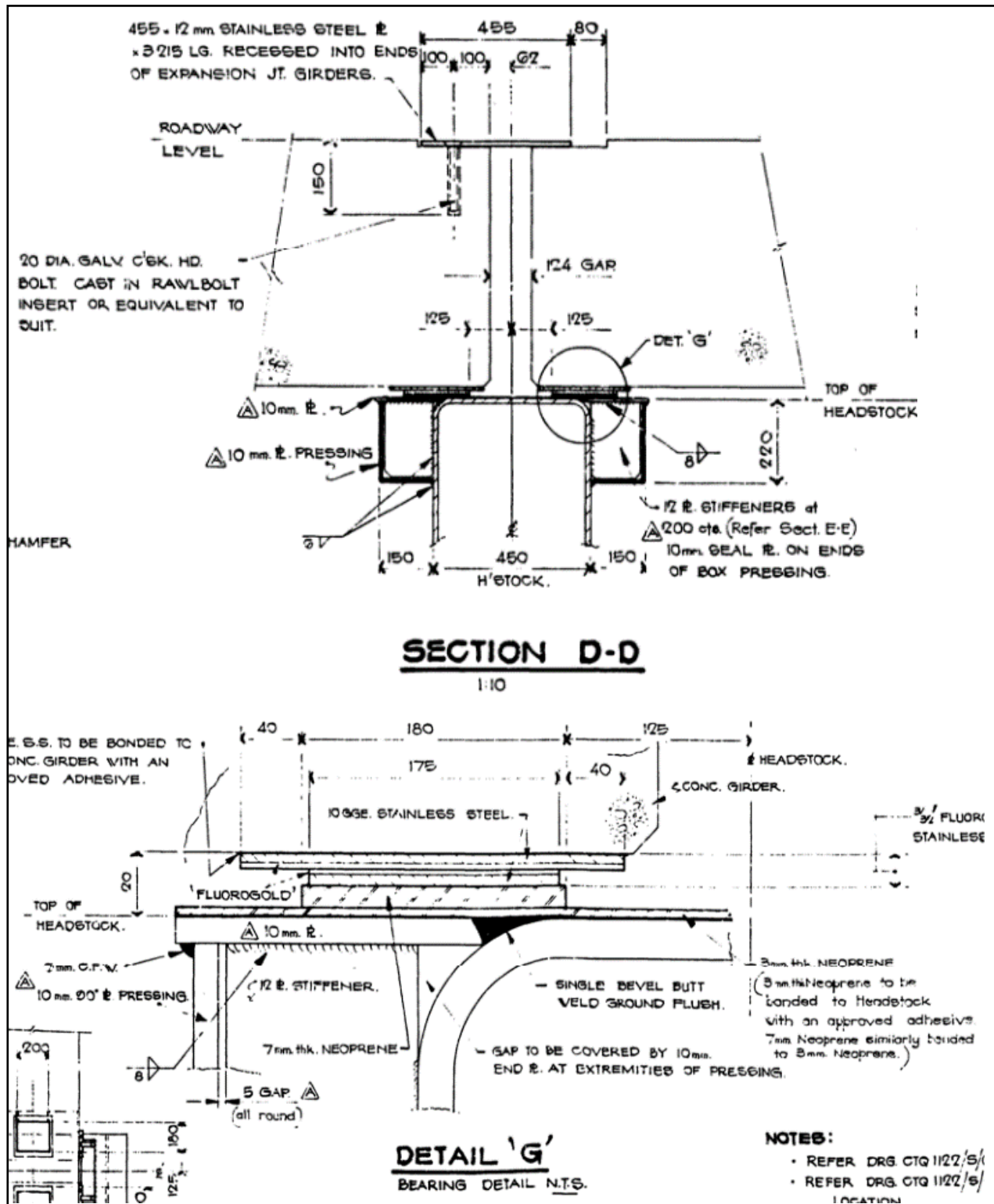


Figure 6: Extract of drawing “Approach Jetty Expansion Joint Details” (Drawing No. CTQ1122/S/032) showing expansion joint and bearing detail

Based on documents provided by QSL, in 1997 / 98, 6no. of the 26no. expansion joints headstocks were inspected and repaired. In addition, it is understood that Bent 280 was also repaired as part of the Cyclone Yasi repairs in 2012/13. The deck units were jacked up using a lifting frame as shown in Figure 7. It is understood that the top surface of the headstocks were found to be in very poor condition with corrosion, pitting and holes evident.

In general, welding repairs were required to the top surface of the headstocks which involved welding a ~3mm thick stainless steel plate with "Flurogold" bearings bonded onto the plate for each PRC deck unit.



Figure 7: Photo showing lifting frame used to lift the deck units on a expansion joint headstock

4.4 Wharf Piles – Current Condition Summary

Based on the SCMC 2022 WSCAM condition assessment data, 107 of the 159 wharf piles have already been wrapped. Figure 8 shows the 52 piles that have not yet been wrapped with the colour of the circle corresponding to the WSCAM CS rating (based on 2022 SCMC condition assessment).

The blue circles show the 13 piles that have already been wrapped but require a minor repair due to localised puncture or missing flange bolts in the pile wrap system.

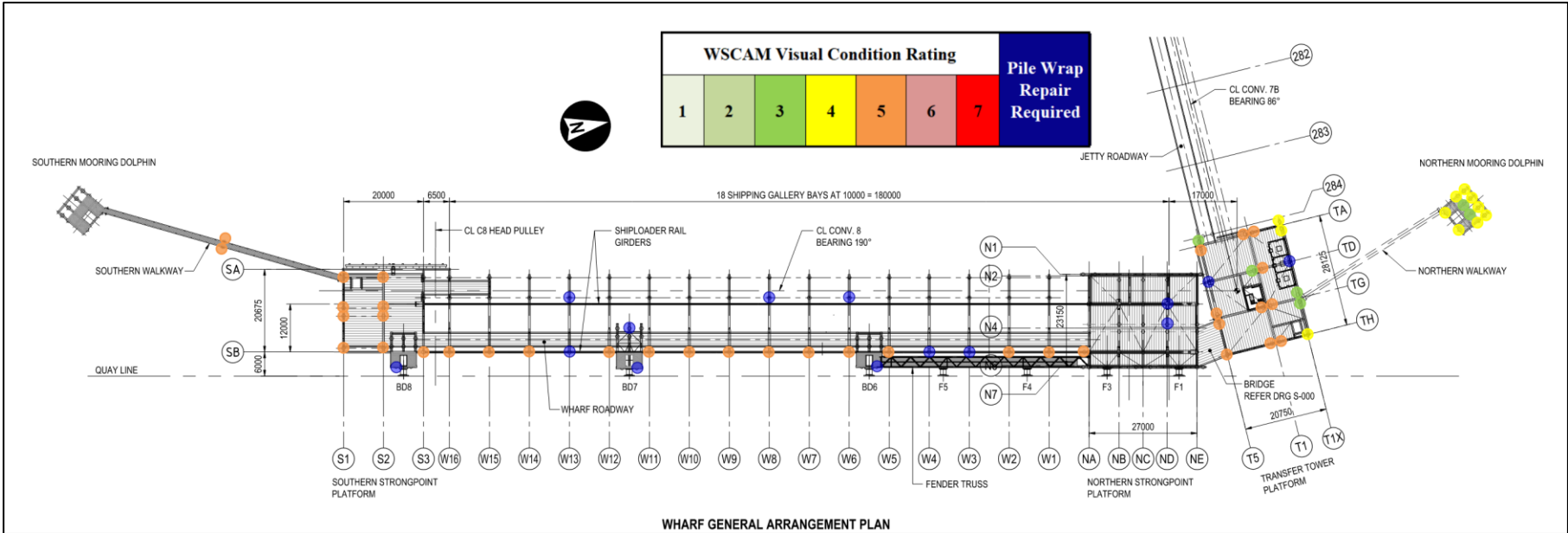


Figure 8: WSCAM heat map of wharf piles that require maintenance repairs. Note: Piles that require minor wrap repair (wrap punctures or missing flange bolts) are shown in blue.

4.5 Wharf Headstocks – Current Condition Summary

Based on the SCMC 2022 WSCAM condition assessment data, Table 5 below shows the number of headstocks within each WSCAM CS rating category. Of the 200 headstocks on the wharf, 67 appeared to have been recently recoated with no defects observed and therefore unlikely to require any maintenance works.

Maintenance repair works are recommended on the remaining 133 headstocks and these headstocks have a WSCAM CS rating between 2 and 5 as illustrated in Figure 9.

Table 5: Number of wharf headstocks with respect to WSCAM CS Ratings

Wharf Structure	1	2	3	4	5	6	7
Southern Mooring Dolphin			7				
South Dolphin Walkway			2	1			
Southern Strongpoint Platform			2	5	4		
Wharf Roadway		14	29	14	10		
Berthing Dolphin 6			9				
Berthing Dolphin 7			5				
Berthing Dolphin 8			9				
Northern Strongpoint Platform		49					
Transfer Tower Platform		1	12	9	6		
North Mooring Dolphin			6	1			
Fender 1		1					
Fender 2		1					
Fender 3		1					

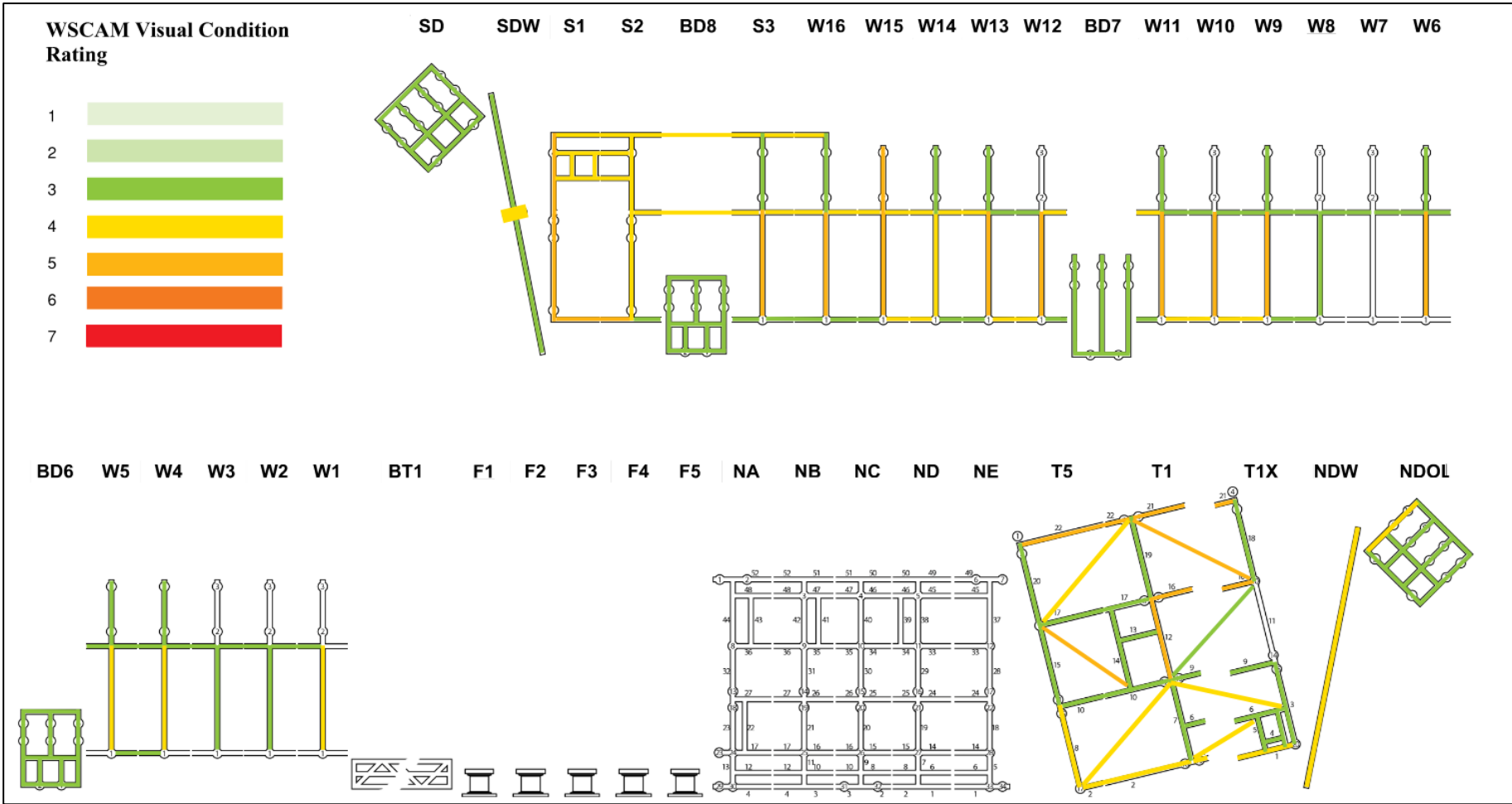


Figure 9: WSCAM CS heat map of wharf headstocks that require maintenance repairs

5 LBST Maintenance Strategy Options

5.1 Proposed Maintenance Strategies

This section of the report discusses the proposed maintenance strategies to be adopted and assessed in Phase 2 of the project - Development of the Financial Model.

SCMC believe that the primary production constraint with the current repair methodology is the JMV as discussed in Section 3.2. SCMC therefore recommend that the focus and direction of the Phase 2 financial model should be assessing options that will speed up the production rate of the repairs, specifically by looking at various options with multiple repair work fronts.

Accordingly, the maintenance strategies proposed for further investigation in Phase 2 are as follows:

1. Baseline Strategy (Current repair methodology):

- a) **Jetty:** Comprising one work front (utilising existing JMV access) working on CS 5 condition headstocks & piles to best (CS 3).
- b) **Wharf:** Undertake repair works in zones based on priority as discussed in Section 3.2.

2. Three (3) Repair Work Fronts Strategy:

- a) **Jetty** - 2 work fronts working separately in parallel:
 1. Pile repairs - one work front (utilising existing JMV access) working on CS 5 condition piles to best (CS 3);
 2. Headstock repairs - one work front (scaffold access) working on CS 5 condition headstocks to best (CS 3);
- b) **Wharf** – 1 work front:
 3. Separate work front undertaking repair works in zones based on priority as discussed in Section 3.2.

3. Four (4) Repair Work Fronts Strategy:

- a) **Jetty** - 3 work fronts working separately in parallel:
 1. Pile repairs - one work front (utilising existing JMV access) working on CS 5 condition piles to best (CS 3);
 2. Headstock repairs - Work front No. 1 (scaffold access) working on CS 5 condition headstocks to best (CS 3);
 3. Headstock repairs - Work front No. 2 (scaffold access) working on CS 5 condition headstocks to best (CS 3);

Note, subject to confirmation of feasibility by QSL's nominated contractor, it may be possible to combine some pile repairs with the headstock repairs using the scaffold platforms;
- c) **Wharf** – 1 work front:
 4. Separate work front undertaking repair works in zones based on priority as discussed in Section 3.2.

4. Five (5) Repair Work Fronts Strategy:

a) **Jetty** - 4 work fronts working separately in parallel:

1. Pile repairs - Work front No. 1 (utilising existing JMV access) working on CS 5 condition piles to best (CS 3);
2. Pile repairs - Work front No. 2 (procure and utilise a second JMV (estimate 2 year delay in commencement of repair works) working on CS 5 condition piles to best (CS 3);
3. Headstock repairs - Work front No. 1 (scaffold access) working on CS 5 condition headstocks to best (CS 3);
4. Headstock repairs - Work front No. 2 (scaffold access) working on CS 5 condition headstocks to best (CS 3);

Note, subject to confirmation of feasibility by QSL's nominated contractor, it may be possible to combine some pile repairs with the headstock repairs using the scaffold platforms;

b) **Wharf** – 1 work front:

5. Separate work front undertaking repair works in zones based on priority as discussed in Section 3.2.

5.2 Discussion

Further to the discussion in Section 3.2, it is recommended that the access method for repair works on the wharf is to assemble a scaffold platform underneath the wharf on an area-by-area basis (i.e. Transfer Tower, Wharf Roadway, Southern Platform, etc.). It is expected that a significant proportion of the repair cost is the assembly and disassembly of the scaffolding and will involve all pile and headstock repairs once the scaffolding is setup in each area.

For all of the proposed maintenance strategies, the JMV is to be used solely to continue the pile repairs along the jetty. Option 5 also involves / includes the proposed procurement of an additional JMV with the strategy to continue using the existing JMV in parallel with the new unit and then be retired from service after all jetty pile and headstock repairs are completed.

The assumed access methodology for the jetty headstocks is scaffold platforms as this will enable full encapsulation for the sand blasting works, noting that scaffold access is expensive. However, it is important to highlight that SCMC believes there are multiple opportunities available to QSL to significantly reduce the cost of accessing and repairing the jetty headstocks by either investigating different procurement methodologies for the scaffold platforms and/or fabrication of a bespoke moveable platform. There is also the alternative repair option of wrapping some or all of the headstocks vs. painting and blasting repairs however this would need to be assessed to determine suitability in terms of long term durability, cost, installation and access requirements, suitability for use in headstock to PRC girder interface, etc. If deemed to be suitable / preferred, this could be easily updated within the financial model as discussed in further detail below.

It is also noted that as discussed in Section 4, the majority of the elements requiring repair have a relatively similar condition. For example, ~74% of the jetty piles yet to be wrapped are WSCAM CS rating of 5. Therefore relying on solely using the WSCAM condition rating will have little benefit in determining the repair priority / order.

Regarding prioritisation of the repair works, this will need to be assessed in further detail once the preferred maintenance strategy is determined. In terms of programming out the repair works, there are multiple factors / variables that can be analysed further to assist with the prioritisation / programming of repairs, such as:

- Capacity utilization of members,
- Area of corrosion on members,
- Location i.e. grouping of adjacent headstocks/piles.

For all of the above options, the financial model will use the fixed unit rate repair costs as provided by QSL and AAB for the following repairs:

- Jetty Piles:
 - Weld repair (\$/m2)

- Pile wrap (\$/pile)
- Jetty Headstock (typical):
 - Weld Repair (\$/m2)
 - Blast clean and reapply protective coating (\$/headstock type based on pile bent width) or wrap repair
- Jetty Expansion Joint Headstock:
 - Jacking of concrete deck units, weld repair, bearing replacement and last clean and reapply protective coating (\$/ expansion joint, to be confirmed)
- Wharf Piles:
 - Weld repair (\$/m2)
 - Pile wrap (\$/pile)
- Wharf Headstock:
 - Weld Repair (\$/m2)
 - Blast clean and reapply protective coating (\$/headstock length / type) or wrap repair

5.3 Potential Opportunities for Accelerating Maintenance Repairs

As discussed between QSL, AAB and SCMC in the workshop, a variety of potential opportunities to accelerate and optimize the maintenance repairs were identified and are listed below:

- Optimising the usage of JMV resource:
 - Options for working smarter (improved resourcing, correct tools and processes);
 - Working longer shifts, spilt shifts and/or weekends;
 - Working out how to best interact with tides i.e. neap vs. spring tide cycles, etc.
- Procurement of an additional (secondary) JMV. The existing JMV could then be retired from service after all pile and headstock repairs are completed.
- Jetty Headstock Access Methodology:
 - Investigate alternative procurement options for scaffolding (vs. subcontracting currently), for example:
 - QSL to procure the required scaffolding and directly employ scaffolding staff (or re-train existing staff)
 - Procurement of a long-term scaffolding contract
 - Manufacture a movable platform. The platform could be moved between headstocks by either:
 - Installing buoyancy units to enable the platform to be floated to the next headstock before it is lifted up and locked off to the headstock
 - Installing the platform using QSL's existing on-site Frana crane positioned on the roadway (potentially in segments)
- Investigate the purchase or hire of a jack up barge.
- Installation of pile jackets (separate to the JMV pile repair works) using a separate team and work-boat/pontoon.
- Review of current structural assessment and weld repair methodology including:
 - Investigation into alternative repair methodologies such as structural pile wraps and pile infilling.
 - Perform a thorough review / detailed structural assessment of the current weld repair strategy to identify potential opportunities to optimise the size and/or number of weld repairs.

- Utilising alternative scanning technologies such as Pulsed Eddy Current Testing (PECT) to efficiently scan large steel surface areas. The key benefit of this testing is that this could allow for a more refined / targeted structural assessment to be performed to minimize the number and / or size of weld repairs required.
- Adopt a risk based approach and investigate the option of accepting a lower structural capacity to reduce the number and / or size of weld repairs required.

It is important to highlight that, in general, whilst these potential opportunities will likely result in probable time and / or cost savings to QSL, they will not necessarily result in a fundamental change in the need for welding repairs, pile wrapping and headstock recoating repairs. Accordingly, further detailed investigation is therefore required to confirm the feasibility, overall project savings and potential upfront costs (e.g. procurement cost of an additional JMV) of the various opportunities identified.

Furthermore, the Phase 2 financial model spreadsheet will be setup such that the input variables for each maintenance strategy can be varied by QSL based on further assessment of the opportunities above, once they have been assessed further and quantified.

Appendix A SCMC Review Comment Register

Refer to attached excel spreadsheet register, “SCMC-22020-REVIEW-001 (REV 1)”

Appendix B Workshop Presentation and Minutes

Refer to attached powerpoint presentation, “22020 QSL LBST Jetty Maintenance Cost Modelling-Meeting Minutes 230207.pptx”